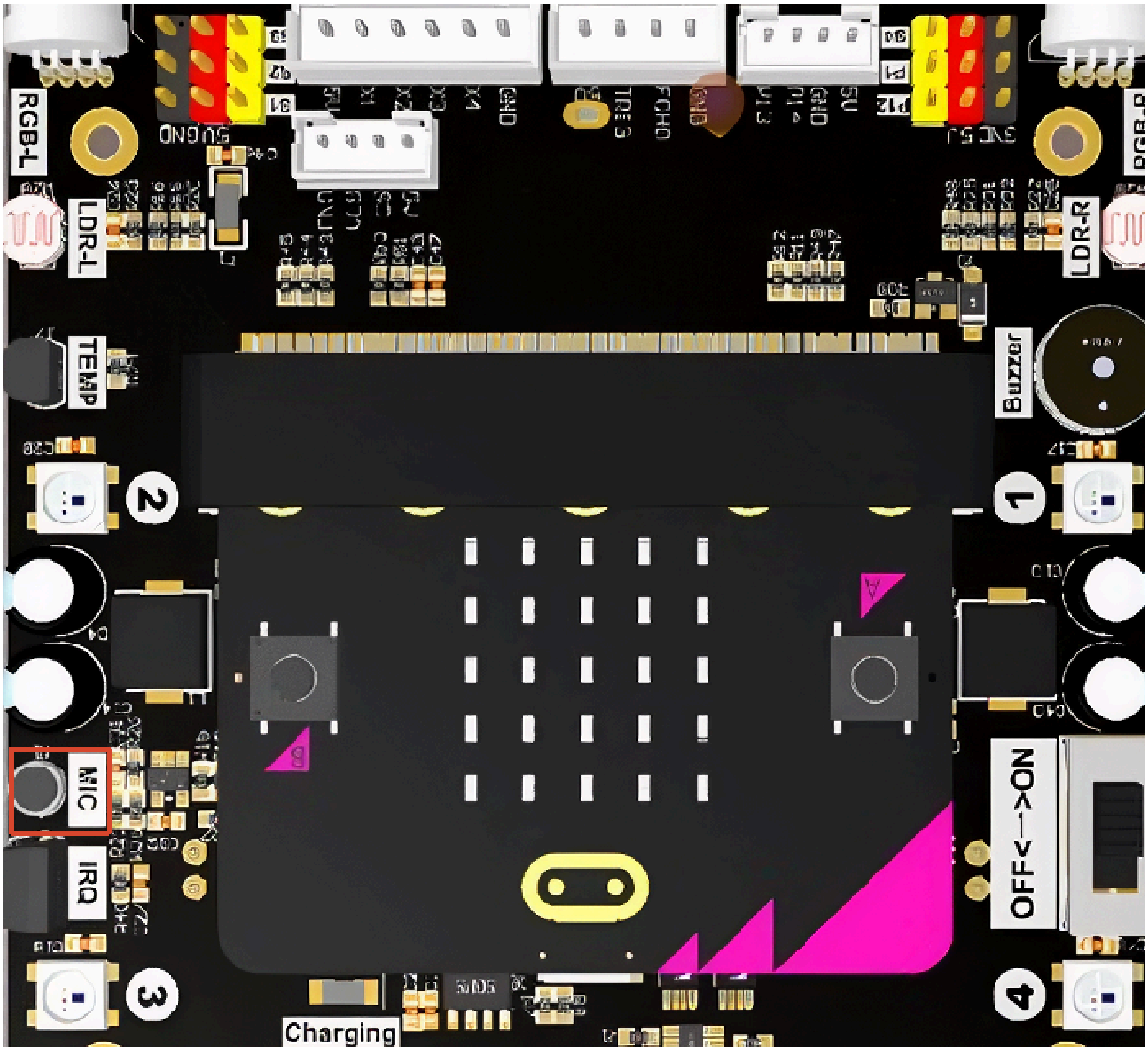
MIC Sound Intensity Collection

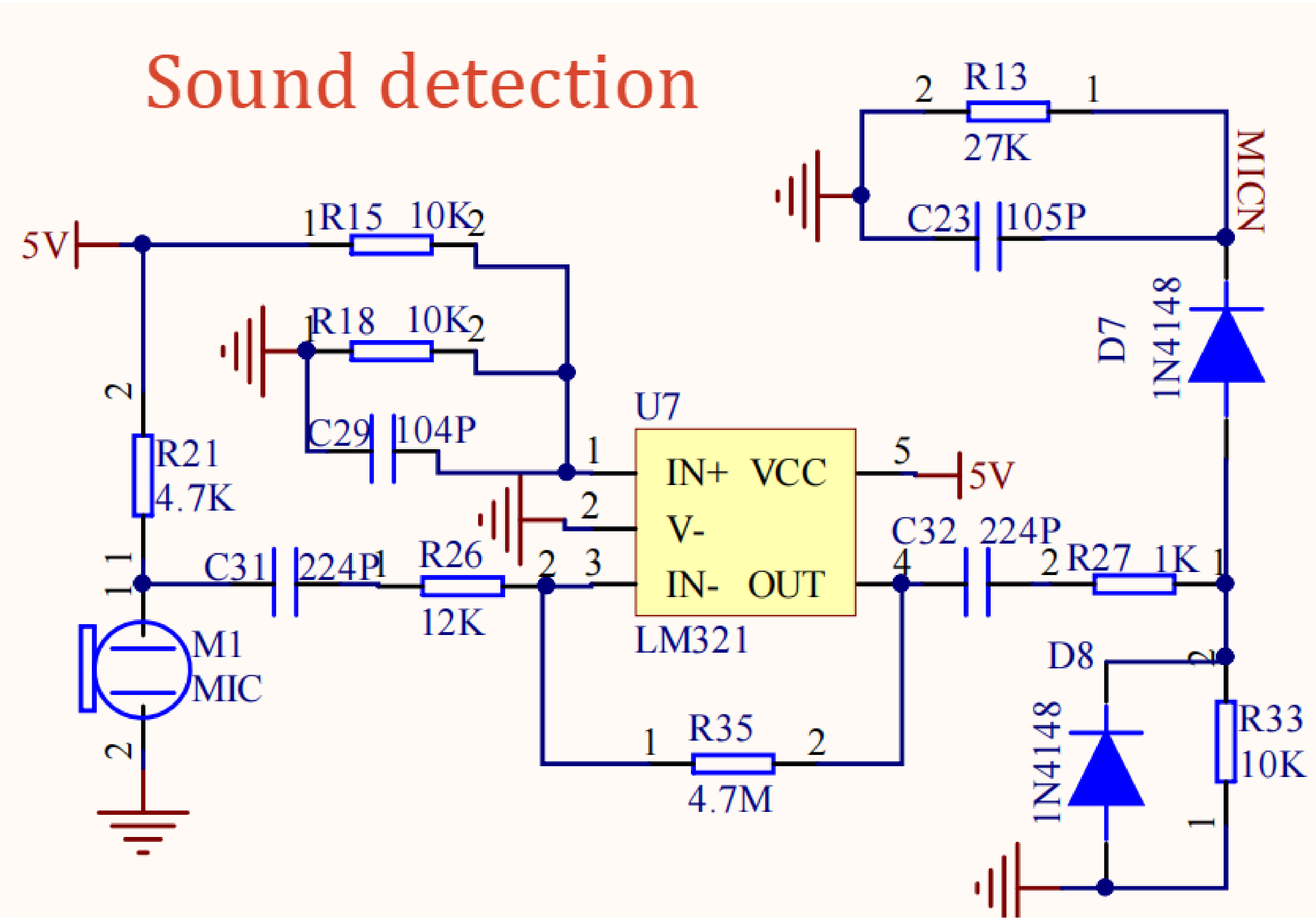
1. Experiment Purpose

The Lastbit car is equipped with a sound sensor on board, with the silk screen marking MIC. This lesson will guide you through learning how to read the intensity information returned by the sound sensor. The larger the value, the noisier the environment the car is currently in.

> **Note**
>
> **The Micro:bit V1.0 version does not have an internal microphone. You can use the MIC on the expansion board to achieve sound detection functionality.**



The onboard sound sensor consists of an electret microphone (commonly called a mic capsule), which forms a complete sound detection circuit through signal amplification and clamping protection.

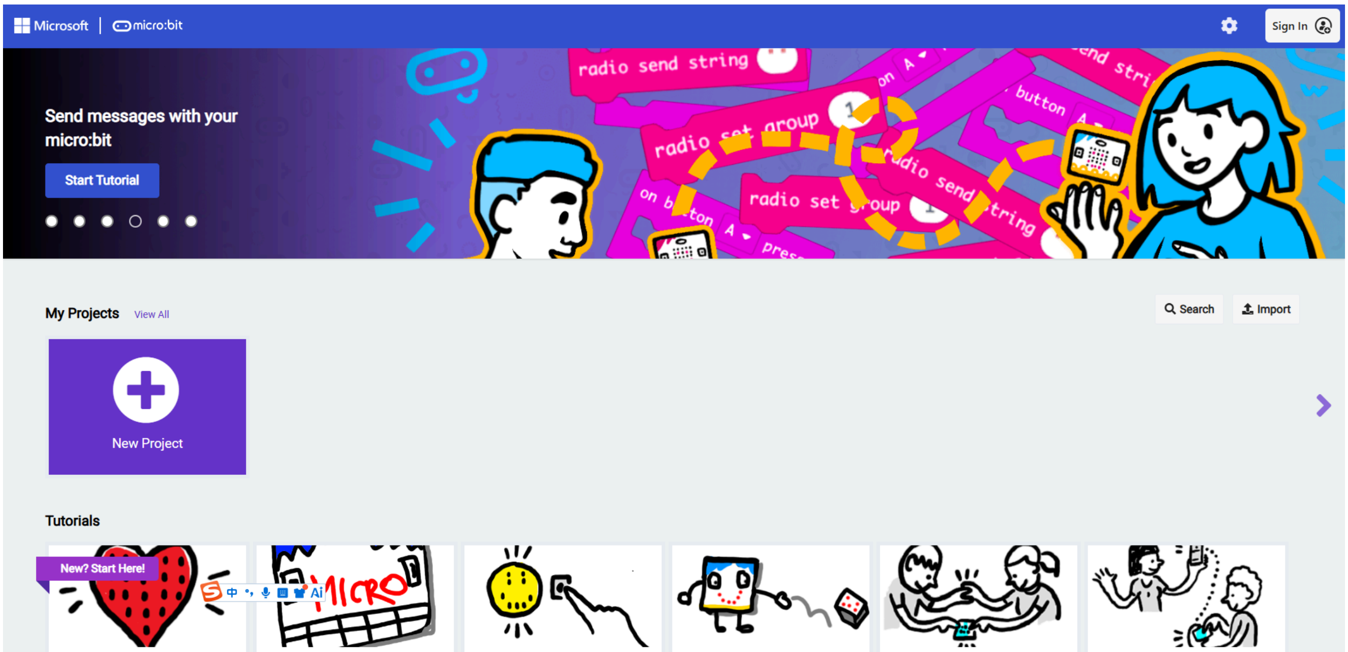


2. Experiment Steps

1. Before starting the experiment, connect the Micro:bit mainboard to your computer using a Micro USB data cable. At this time, the computer will display the MICROBIT drive letter.

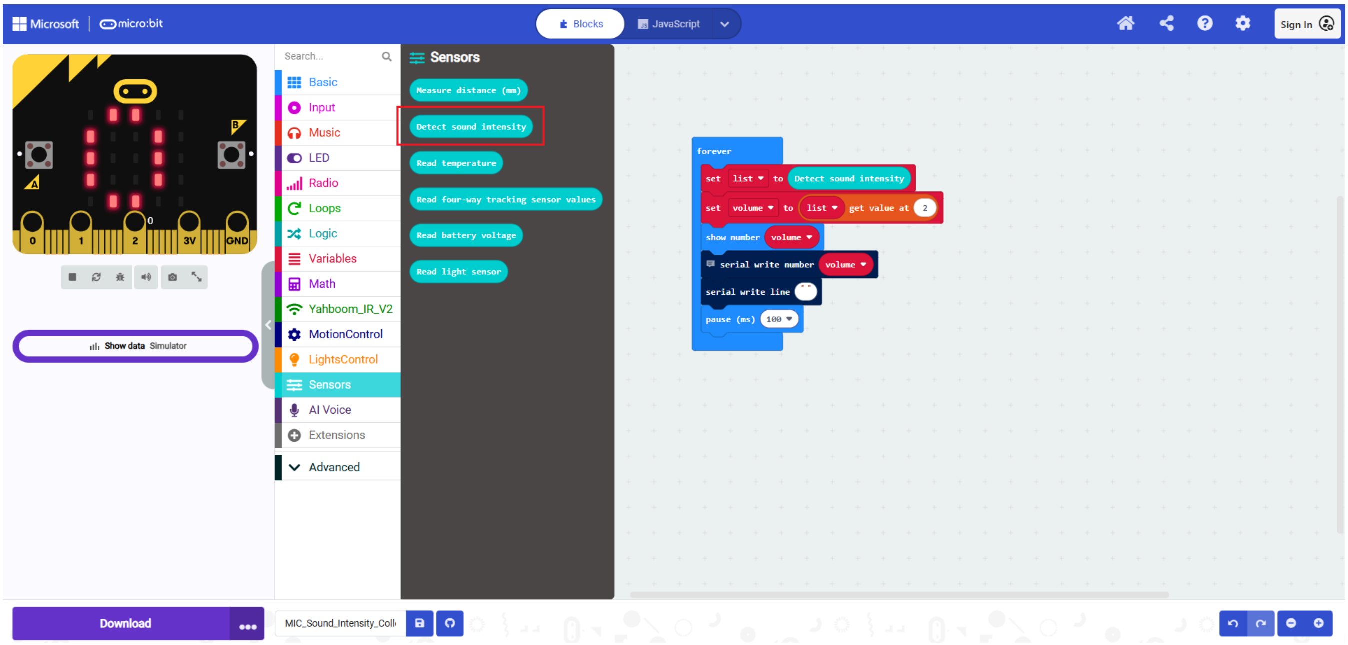
2. Open MakeCode using the link below.

   [Microsoft MakeCode for micro:bit]()



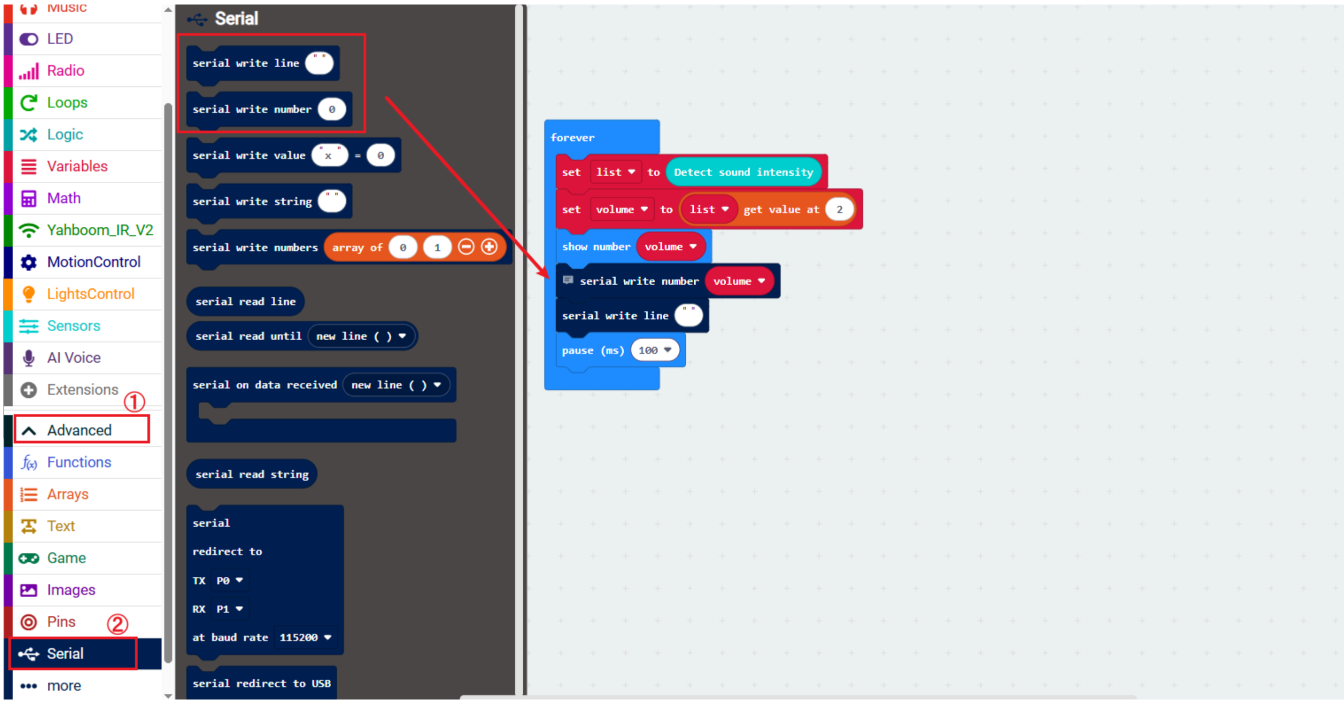
3. Drag `6.1 MIC_Sound_Intensity_Collection_V1V2.hex` to the new project page. MakeCode will automatically import and open the project.

3. Experiment Source Code



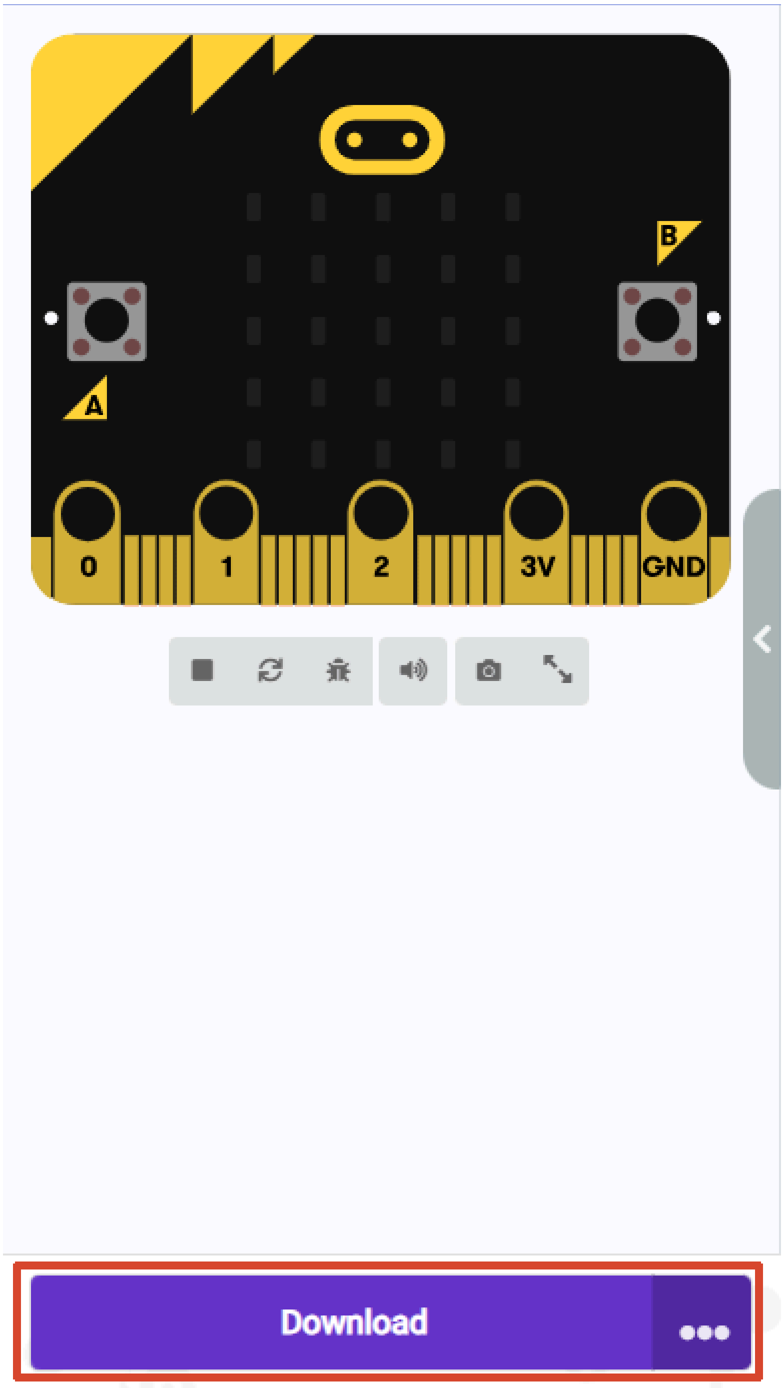
The core building blocks for this lesson are located in the `Sensor` category. The serial building blocks are located in the `Serial` subcategory under the `Advanced` category. The two lines between displaying the number and pause involve the `Serial` building blocks which can be deleted. If a serial assistant is connected, it will print data, which does not affect the sensor data display.

Here is a brief explanation of the index values: Index 0 is the minimum value captured, index 1 is the maximum value captured, and index 2 is the average value captured. The average value is relatively more stable.

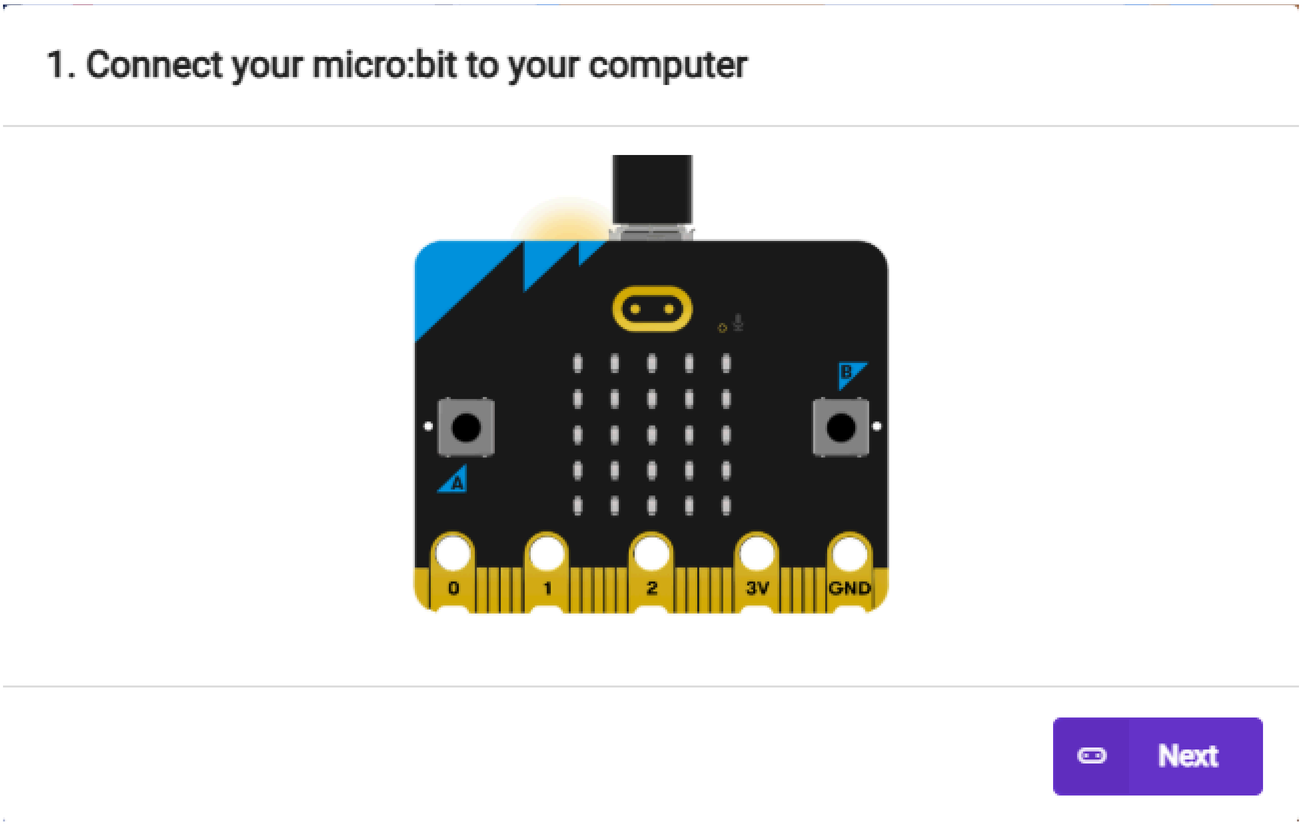


4. Program Download

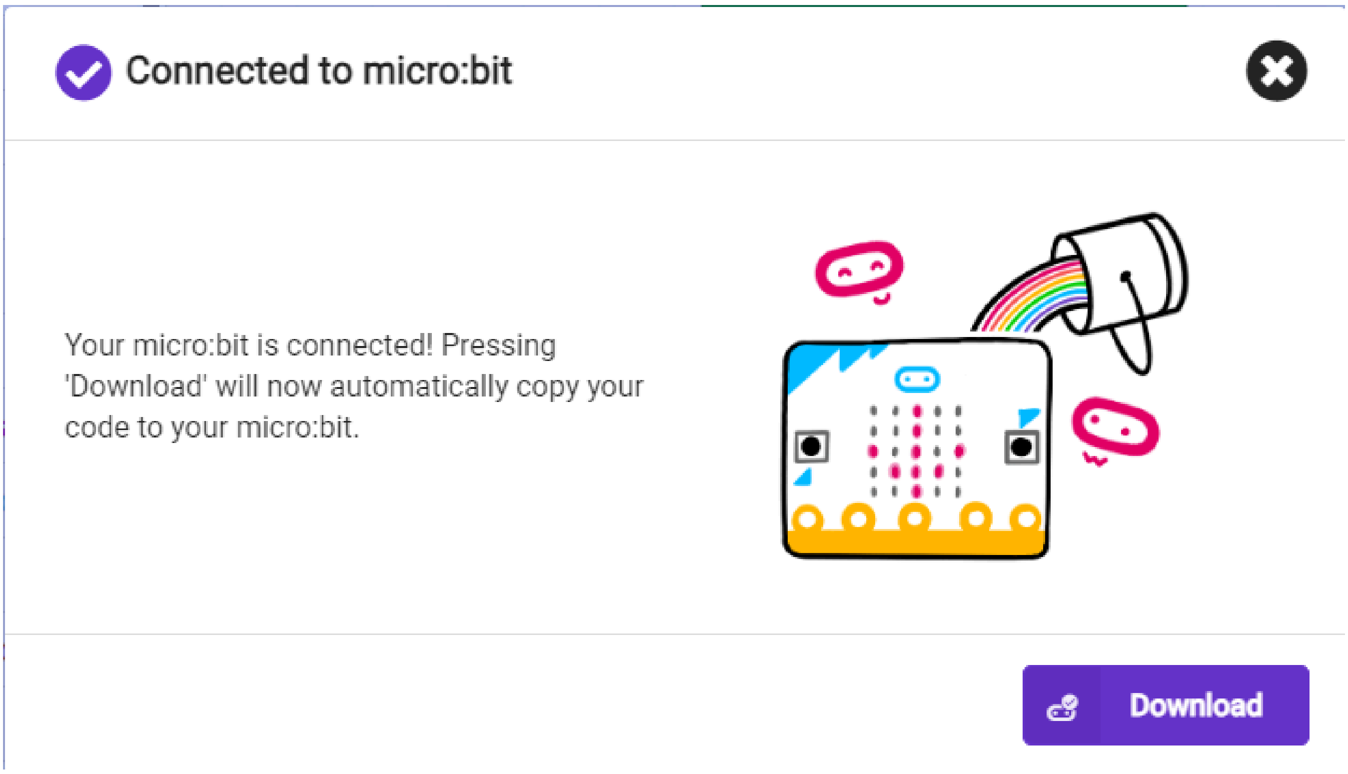
1. Click the download button at the bottom left to download the program to the Micro:bit.



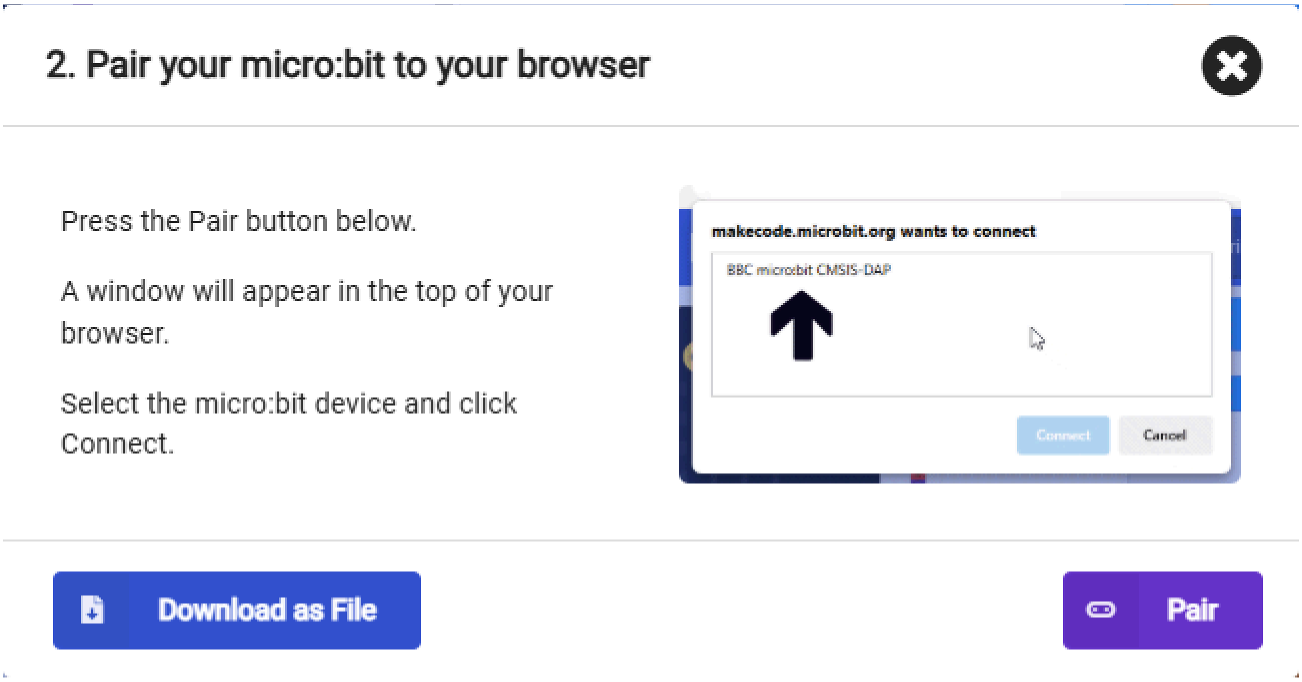
2. When the Micro USB data cable is not connected to the Microbit, the system will first prompt you to connect the device. If not connected, please connect the data cable to the Microbit first. If already connected, click [Next] here.



3. If the Microbit is detected as properly connected, the following interface will be displayed. At this point, you can click [Download].

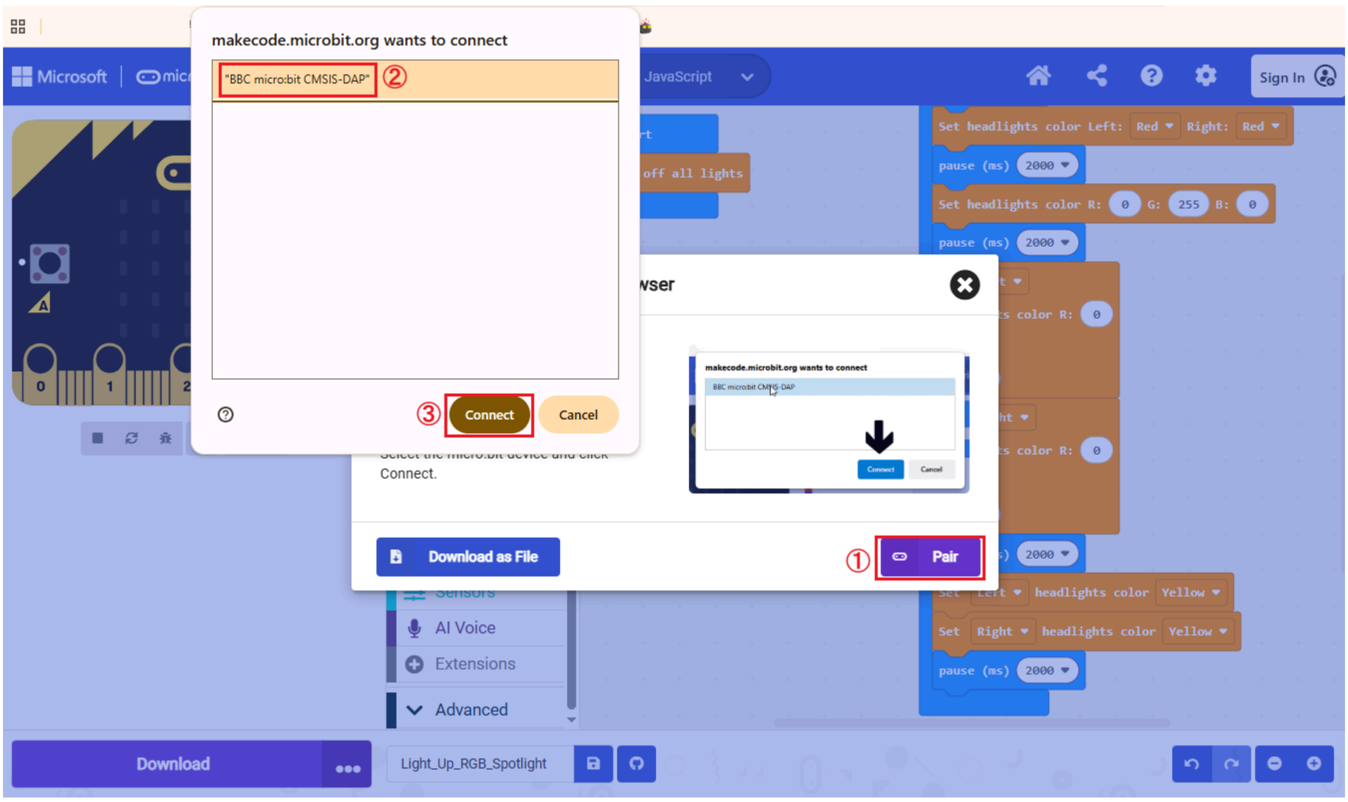


If the Microbit data cable is not properly connected or multiple Microbits are connected, the system will require you to manually select the device. You can choose between two methods: [Download as File] and [Pair].

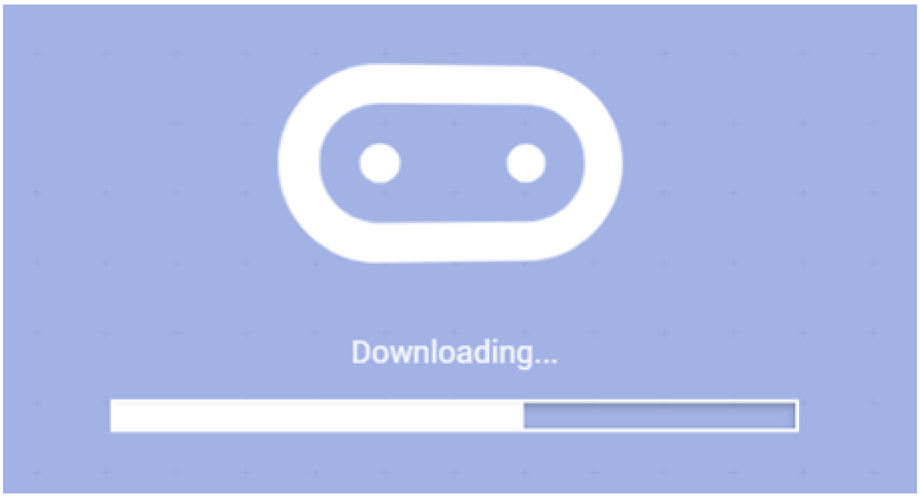


When selecting [Download as File], the system will download the currently opened program as `microbit-project-name.hex`. You can directly copy or drag this program to the corresponding MICROBIT drive letter to complete the download.

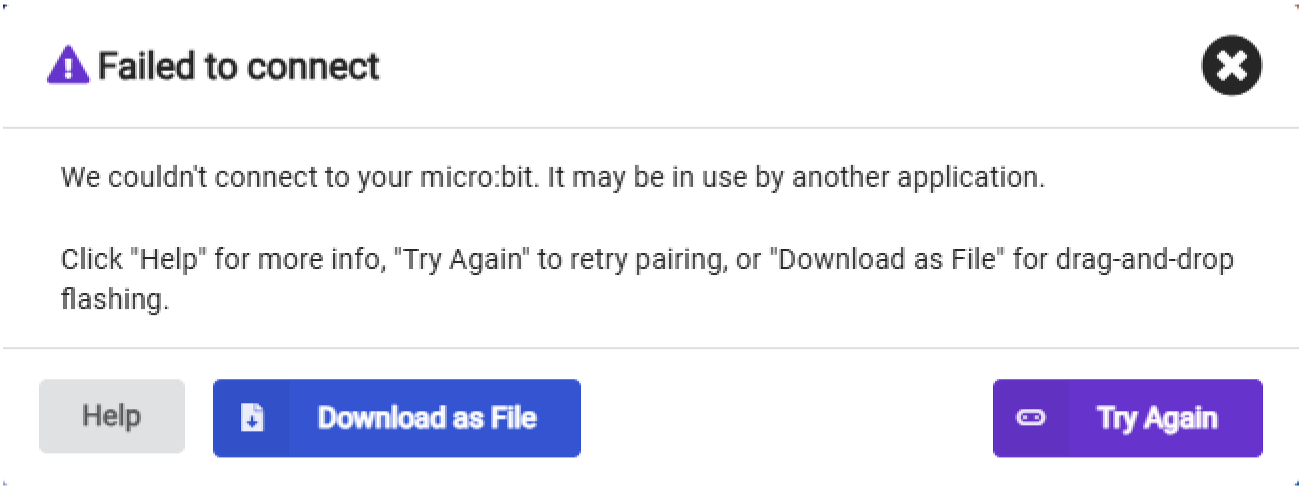
When selecting [Pair], you will be asked to manually select the device. Here, select `BBC mico:bit CMSIS:DAP`.



4. When the following interface appears, it means the program is being downloaded. Wait for the progress bar to complete to download the program to the Microbit.



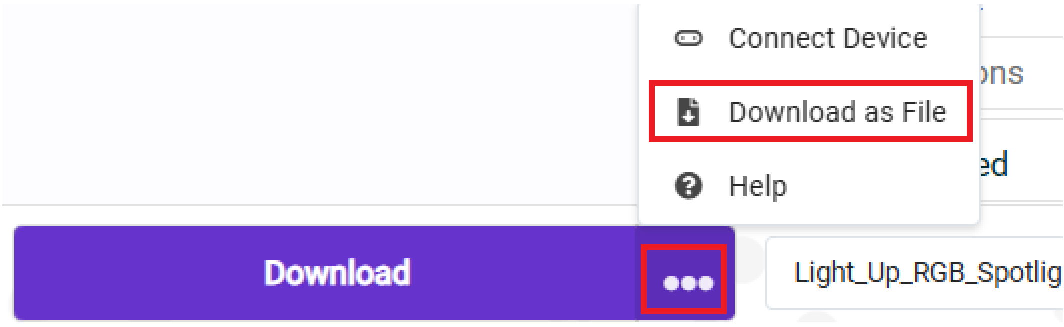
If multiple attempts continue to fail, please directly select [Download as File], save the current program as a file, and directly copy or drag the program to the corresponding MICROBIT drive letter to complete the download.



If you want to save the file directly, we provide two methods:

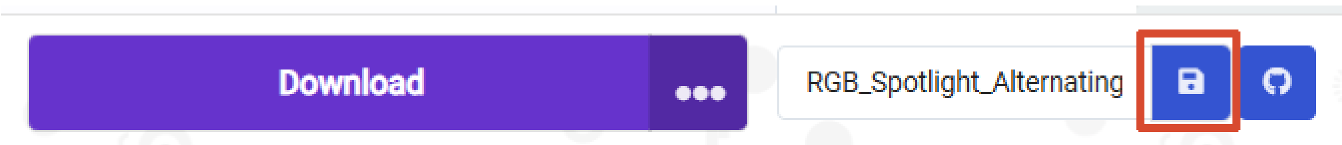
Method 1:

Click the [...] next to download, then select [Download as File]



Method 2:

Click [Save] after the [Project Name Edit Field].



5. Experiment Results

After flashing, turn on the car switch (flip the switch to the ON position). The Micro:bit LED matrix screen will display the sound intensity data acquired by the sound sensor.